Premier Energies

TALENT OPERATIONS

ATS User Guide

A complete, code-aligned manual covering every screen, action, status, email, schema column, and TAT pair.

VERSION	UPDATED	AUDIENCE
2.1 — Phase 0–10 rebuild	April 2026	HR Admin · Recruiter · Approver · Interviewer

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Premier Energies ATS — User Guide

Version: 2.1 (Phase 0–10 rebuild) · **Last updated**: April 2026

A complete, code-aligned manual for the Premier Energies Applicant Tracking System. Every screen, action, status, email, schema column, and TAT pair documented in one place.

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1. Overview & Architecture

The Premier Energies ATS is a single-tenant, role-based hiring platform spanning the full lifecycle: **requisition → approval → job → sourcing → interview → selection → documents → CTC → offer → joining**.

Stack:

- **Server**: Node.js + Express, PostgreSQL via `pg`, JWT auth, Microsoft Graph for outbound email
- **Client**: React 18 + Vite + Tailwind CSS, design-token CSS layer (`v2.css`)
- **AI**: Local Ollama for resume parsing + analytics copilot (graceful fallback if unavailable)
- **Reminders**: Setinterval-based poll runner (`reminders.js`) — no external job queue
- **Brand**: `pel.png` logo + `l.png` wordmark, served at `/pel.png` and `/l.png`

Top-level project layout:

```
ats/
├─ server/
│  ├─ index.js              ← Express bootstrap
│  ├─ middleware/auth.js    ← JWT + role gates
│  ├─ migrations/run.js     ← Idempotent schema (npm run migrate)
│  ├─ routes/               ← One file per endpoint family
│  └─ services/             ← Email, AI, TAT, audit, timeline, reminders
└─ client/
   ├─ src/
   │  ├─ pages/             ← Top-level routes (28 pages)
   │  ├─ components/        ← Shared widgets (TriageMenu, SignaturePad, …)
   │  ├─ components/ui/     ← Design system primitives
   │  ├─ services/api.js    ← Axios + endpoint families
   │  └─ styles/            ← index.css + v2.css token layers
```

2. Roles & Access Control

Five built-in roles, defined in `users.role`:

ROLE	CAPABILITIES
`hr_admin`	Everything: masters, users, all approvals, all data, audit deck, demo seed, blacklist lift
`hr_recruiter`	Sourcing, triage, interviews, document reviews scoped to own assignments, CTC drafting
`hod`	Raise requisitions, approve at HOD/CXO stage when listed in `approvers_master`, view assigned candidates
`interviewer`	View assigned candidates, suggest slots, submit feedback, request more rounds
`applicant`	Candidate Portal — chat, document uploads, CTC accept, offer signature

Login: passwordless OTP. `POST /api/auth/login` issues a 6-digit code expiring in 5 minutes via Graph email; `POST /api/auth/verify-otp` exchanges it for a JWT. Logout calls `POST /api/auth/logout` which writes a `session/delete` audit row.

Default admin: created on first migration with `is_default = true`. Cannot be deactivated or have role changed.

Role gates (defined in `middleware/auth.js`):

```
requireRole('hr_admin')               // strict admin only
requireRole('hr_admin', 'hr_recruiter') // admin OR recruiter
```

3. End-to-End Hiring Flow

```
[ Requisition ] → [ Approval ] → [ Job ] → [ Sourcing ] → [ Triage ]
        ↓             ↓             ↓                          ↓
   created_by    cxo, hr_admin  recruiter assigned    public apply or upload

[ Triage ] → [ Shortlist ] → [ Interview rounds ] → [ Selected ]
                  ↓                ↓                        ↓
            override rounds   schedule, no-show,    documents loop
                              more rounds

[ Documents cleared ] → [ CTC chain ] → [ CTC accepted ] → [ Offer letter ]
                            ↓                ↓                      ↓
                      recruiter1 → 2 →   candidate accepts    digital signature
                      hr_admin →
                      optional approver

[ Offer signed ] → [ Joining date ] → [ Joined / Postpone / Dropout ]
```

3.1 Requisition raise

1. Navigate to `/requisitions/create`
2. Pick **type**: `new_hire` (routes through CXO + HR Admin) or `replacement` (HR Admin only)
3. Fill BU/Department/Sub-Department/Location/Phase/Grade/Level + position table
4. Submit:
   - Status set by `resolveSubmissionStatus()` based on type + role
   - `requisition_approvals` rows created via `replaceApprovalSteps()`
   - **Notifications**: admin alert email (`requisitionRaisedAdminEmail`), raiser confirmation email (`requisitionRaisedConfirmationEmail`), in-app to all approvers

3.2 Approval

- CXO approver acts via `POST /api/requisitions/:id/approve`
- After all CXO clear → status moves to `pending_hr_admin_approval` and HR Admin approval steps are inserted
- HR Admin approves → `status = approved`, `approved_at` set
- Either side can `POST /:id/reject` with comments → `rejected` or `cxo_rejected`

3.3 Job creation

After approval, HR Admin chooses one of three paths from the Requisition Detail page:

PATH	RESULT
Create Job	Opens `/jobs/create` prefilled with the requisition; recruiter can add `hr_one_job_id` + interviewer rounds. Notifies recruiter + admin + raiser on save.
Assign Recruiter(s)	`POST /:id/assign-recruiter` — writes `requisitions.assigned_recruiter_email`, emails them.
Hold with reason	`POST /api/requisition-holds` — TAT pause begins, hold reason emailed to raiser + recruiters.

3.4 Sourcing — application intake

Two intake paths:

PATH	INITIAL STATUS	NOTES
Self-apply via `/careers/:jobId`	`InQueue`	Confirmation email sent (`applicationReceivedEmail`); recruiter notified
Recruiter upload	`Applied`	Source captured from upload form (LinkedIn / Naukri / Referral / etc.)

Public apply gate (`server/routes/public.js`):

- Job must be `active_flag=true` AND status not in `closed/archived/on_hold/filled/cancelled`
- Phone is checked against `blacklisted_phones`; banned applicants get a generic 403 message
- Resume upload uses crypto-random filename so concurrent submissions can't collide

3.5 Triage actions

Available via `<TriageMenu>` on every candidate row (Job Detail, Talent Pool, Application Workflow):

ACTION	ENDPOINT	SIDE EFFECTS
Shortlist	`POST /api/triage/:id/shortlist`	Status → `AwaitingHODResponse`. Optional per-candidate rounds + interviewer override saved to `application_round_overrides` with provenance. Round-1 interviewers emailed.
Move to job	`POST /api/triage/:id/move-to-job`	Status → `Applied` in target job. Searchable picker shows full label including HR One ID.
Move to talent pool	`POST /api/triage/:id/move-to-talent-pool`	`ats_job_id='TP-POOL'`, status `TalentPool`. Movement row in `talent_pool_movements` retains from-job + reason + actor.
HR reject	`POST /api/triage/:id/hr-reject`	Status → `HRRejected`. Goldman-style polite decline email auto-sent (`politeDeclineEmail`).
Blacklist	`POST /api/blacklist`	Phone banned in `blacklisted_phones`. Status → `Blacklisted`. Admins emailed with resume attached as Graph base64 attachment.

3.6 Interview flow

1. **Shortlisted** → `AwaitingHODResponse` notifies first-round interviewers
2. Interviewer reviews via `/interviews` and either:
   - **Suggests slots** → status → `AwaitingInterviewScheduling`
   - **Rejects** with reason → `HODRejected` (option to retain in talent pool)
3. Recruiter confirms a slot OR overrides with own time picker → status → `Round1`
4. After interview + 10-minute buffer:
   - **No-show** allowed (425 returned if too early). Candidate gets the empathic 24h grace email (`noShowGraceEmail`). Application returns to `AwaitingInterviewScheduling`.
   - **Feedback submission** with technical/behavioral/fit scores + decision:
     - `shortlist` (mid-rounds) → next round, or → `Selected` if final round
     - `reject` → `Round{N}Rejected`
     - `request additional rounds` → adds rounds with new interviewers + suggested slots, status → `AwaitingInterviewScheduling`
5. Reminder runner (`reminders.js`) auto-schedules T-24h and T-30m candidate emails when a slot is confirmed

3.7 Documents loop

Triggered when status reaches `Selected`. Default document set is seeded by `ensureDefaultSelectedDocuments()`:

STAGE	DOCUMENTS
`before_offer_release`	PAN, Aadhaar, Latest payslip / Bank statement, Offer letter / Employment proof
`before_joining`	(recruiter adds custom requests)
`joining_day`	(recruiter adds custom requests)

Loop:

1. HR adds documents/questions via Document Review Queue
2. Candidate uploads via Candidate Portal (`POST /api/candidate-portal/documents/:docId/upload`)
3. Recruiter reviews via `/hr/document-queue`:
   - **Accept** → status `accepted`
   - **Reject with comment** → status `rejected`, candidate notified to re-upload
4. Candidate re-uploads → version bumps; cycle continues
5. Once all documents `accepted` → status moves to `DocumentsCleared` (recruiter manually advances)

3.8 CTC chain

Triggered after documents cleared. 2–4 step chain stored in `ctc_approval_chain`:

```
[Recruiter 1 drafts] → [Optional Recruiter 2] → [HR Admin] → [Optional Approver]
       ↓                        ↓                    ↓                ↓
  approve / reject /     same options          same options    same options
   renegotiate
```

Each step can **approve** (forward), **reject** (`SalaryRejected`, candidate returns to TP), or **renegotiate** (back to Recruiter 1). On final approval, candidate gets `ctcAcceptanceEmailV2` and status moves to `CTCAcceptance` (14-day validity).

3.9 Offer + signature + joining

1. After candidate accepts CTC (`ctc.accepted`), recruiter uploads offer letter PDF (`POST /api/offers/:id/upload`)
   - Default validity 14 days
   - Auto-schedules T-3d and T-1d reminder emails to candidate
   - Status → `SignaturePending`
2. Candidate views offer + signs digitally on `<SignaturePad>` canvas:
   - Captures base64 PNG signature, IP, UTC timestamp
   - Status → `OfferAccepted` (or `OfferRejected` / `OfferDropout` if declined)
   - Stakeholders emailed (recruiter, secondary, requisition creator, admin)
3. Recruiter sets tentative joining date (`POST /:id/joining`)
   - Schedules joining-day reminder for 09:00 IST
4. On joining day, recruiter records outcome (`POST /:id/joining-outcome`):
   - `joined` → status `Joined`
   - `postpone` → status `Postponed` with new date + reason
   - `dropout` → status `OfferDropout` with dropout reason

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4. Pages Reference

4.1 / — Dashboard

Hero + KPIs (open jobs, applications, offer acceptance, avg time to fill) + funnel + recruiter momentum + needs-attention queue. Demo seeding controls for HR Admin.

4.2 /requisitions — Requisitions

List with status filters. Statuses: `draft / pending_approval / pending_cxo_approval / pending_hr_admin_approval / approved / rejected / cxo_rejected / closed / on_hold`.

4.3 /requisitions/create & /:id/edit — Create Requisition

Wizard: type → BU/Dept → Position table (per-row hire window + backfill data) → Approvers preview → Submit.

4.4 /requisitions/:id — Requisition Detail

Tabs: Overview, Approvals timeline, Holds, Audit. Approve/Reject buttons gate by role + pending step ownership.

4.5 /jobs — Jobs

Card and table view. Filters by status / BU / department. Bulk export with/without applicants via `/api/mis/jobs-export?include_applicants=true|false`.

4.6 /jobs/create & /jobs/:id/edit — Create Job

Multi-step wizard: Identity → Compensation → Description → Visibility → Interview Design (rounds + per-round interviewer chips). New `hr_one_job_id` field on Visibility step.

4.7 /jobs/:id — Job Detail

Three tabs:

- **Workflow Queue** — sticky search bar, full-width table with `<TriageMenu>` on every row
- **Pipeline View** — kanban with 5 lanes (Sourcing → Screening → Interview → Selection → Onboarding)
- **Job Settings** — full-width with Identity / Compensation / Description sections + Hold with reason CTA

4.8 /talent-pool — Talent Pool

Candidates with `ats_job_id='TP-POOL'`. Each card shows the movement history ("Moved here from ROLE-XYZ by alice@... on Apr 12").

4.9 /applications/:id/workflow — Application Workflow

Per-candidate command centre. Timeline rail, document checklist, CTC chain modal trigger, interview history, chat trigger.

4.10 /interviews — Interview Hub

Interviewer-facing list of assigned candidates. Suggest slots, submit feedback, request more rounds, mark no-show.

4.11 /candidate — Candidate Portal (applicants only)

Hero + offer signature panel + chat panel + document upload cards by stage + CTC acceptance section.

4.12 /hr/document-queue — Document Review Queue

Scoped to recruiter's own applications (full visibility for HR Admin). Approve / Reject with comment.

4.13 /users — User Management

Active + Inactive lists (Masters-style split). Applicants shown in their own table. Reactivate from inactive list.

4.14 /masters — Masters

Tabs for BUs, Departments, Sub-Departments, Locations, Phases, Grades, Levels, Sources, Approvers Master, Reasons.

4.15 /aop — AOP Dashboard

Annual Operating Plan headcount tracking by BU + grade.

4.16 /mis — MIS Reports

Multi-tab analytics: Funnel, Time-to-fill, Recruiter momentum, Source mix, Demographics, Detailed open positions, Raw export. Filters: date range, recruiter, **talent_pool (include / exclude / only)**, **hr_one_job_id**.

4.17 /tat — TAT Explorer

Hero + level toggle (Requisition / Job / Application) + pair picker + KPI tiles (P50, P90, missing) + full table + **"Show working"** modal that displays the actual rows used to compute each duration plus raw JSON.

4.18 /audit — Audit Deck

Thread-grouped cards. Filters by action type, entity, actor, date range, field-search. Stats KPIs. JSON export.

4.19 /notifications — Notifications inbox

In-app notifications with mark-read, mark-all-read.

4.20 /analytics-copilot — Analytics Copilot

Natural-language Q&A over MIS data. Falls back to data-driven summary if Ollama unavailable.

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5. Application Status Map

All statuses defined in `server/services/interviewWorkflow.js`. Total: **31 statuses**.

STATUS	MEANING	REACHABLE FROM	REACHABLE TO
`InQueue`	Self-applied, awaiting first triage	(initial)	Applied, Shortlisted, HRRejected, Withdrawn, Blacklisted, TalentPool
`Applied`	Recruiter-uploaded or moved-from-other-job	InQueue, (initial)	Shortlisted, HRRejected, Withdrawn, Blacklisted, TalentPool
`Shortlisted`	Triaged in, awaiting HOD acknowledge	Applied	AwaitingHODResponse, HRRejected, Withdrawn, TalentPool, Blacklisted
`AwaitingHODResponse`	HOD/Interviewer needs to suggest slots or reject	Shortlisted	AwaitingInterviewScheduling, HODRejected, Withdrawn, TalentPool
`AwaitingInterviewScheduling`	Slots suggested, recruiter to confirm	AwaitingHODResponse, post-no-show, post-additional-rounds	Round1/2/3, HRRejected, Withdrawn, TalentPool
`Round1` , `Round2` , `Round3`	Confirmed interview	AwaitingInterviewScheduling, prior round	Round{N}Rejected, AwaitingFeedback, next round, Selected, Withdrawn, TalentPool
`Round{N}Rejected`	Round-specific rejection	Round{N}	TalentPool (re-entry)
`AwaitingFeedback`	Interview occurred, feedback pending	Round{N}	Selected, Round{N}Rejected, AwaitingInterviewScheduling, Withdrawn, TalentPool
`Selected`	Final-round shortlist confirmed	Round{final}, AwaitingFeedback	DocumentsInProgress, CTCSent, OfferInProcess, OfferRejected, Withdrawn, TalentPool
`DocumentsInProgress`	Doc uploads + reviews in flight	Selected	DocumentsCleared, Withdrawn, TalentPool, OfferDropout
`DocumentsCleared`	All required docs accepted	DocumentsInProgress	CTCSent, TalentPool, Withdrawn
`CTCSent`	CTC chain in motion	DocumentsCleared, Selected	CTCAcceptance, CTCAccepted, SalaryRejected, TalentPool, Withdrawn

STATUS	MEANING	REACHABLE FROM	REACHABLE TO
`CTCAcceptance`	Candidate review, 14-day window	CTCSent	CTCAccepted, SalaryRejected, OfferDropout, TalentPool
`CTCAccepted`	Candidate accepted	CTCAcceptance	OfferInProcess, OfferDropout, Withdrawn
`SalaryRejected`	HR Admin / approver rejected the CTC	any CTC step	TalentPool, CTCSent (re-draft)
`OfferInProcess`	Recruiter preparing offer letter	CTCAccepted, Selected	SignaturePending, Offered, OfferRejected, OfferDropout, Withdrawn
`SignaturePending`	Letter uploaded, awaiting candidate signature	OfferInProcess	Offered, OfferRejected, OfferDropout
`Offered`	Candidate signed	SignaturePending, OfferInProcess	OfferAccepted, OfferRejected, OfferDropout
`OfferAccepted`	Explicit acceptance recorded	Offered	Postponed, Joined, OfferDropout
`Postponed`	Joining date pushed	OfferAccepted	Joined, OfferDropout, OfferAccepted
`Joined`	Candidate actually joined (terminal)	OfferAccepted, Postponed	—
`OfferRejected`	Candidate declined offer (terminal)	various	—
`OfferDropout`	No-show / pulled out (terminal)	various	—
`HRRejected`	HR rejected with reason (terminal)	InQueue, Applied, Shortlisted, etc.	—
`HODRejected`	HOD rejected (terminal)	AwaitingHODResponse	TalentPool (re-entry)

STATUS	MEANING	REACHABLE FROM	REACHABLE TO
`Withdrawn`	Candidate withdrew (terminal)	many	—
`Blacklisted`	Phone-banned (terminal)	many	—
`TalentPool`	Parked, full history retained	many	Applied, Shortlisted, Withdrawn, Blacklisted

The transition map is enforced server-side by `assertHrManagedTransition()` in `server/services/interviewWorkflow.js`. **Terminal states** cannot be transitioned out of (except `TalentPool`, `HODRejected`, `Round{N}Rejected`, `SalaryRejected` which allow re-entry into the active pipeline).

6. TAT Engine

Defined in `server/services/tat.js`. **16 named pairs** spanning three levels.

6.1 Pair structure

```
{
  id: 'applied_to_joined',          // stable machine id
  label: 'Applied → Joined (end-to-end)',
  level: 'application',             // requisition | job | application
  description: 'Total candidate journey...',
  from: { source: 'audit_trail', entity_type: 'application', action_type: 'create' },
  to:   { source: 'timeline_events', entity: 'application', event_type: 'application.joined' },
  excludeWhen: ['job.status=on_hold'],  // optional pause conditions
}
```

6.2 Source resolvers

SOURCE	LOOKUP	EXAMPLE MATCH SHAPE
`column`	Reads a column from a single row	`{ table: 'requisitions', column: 'submitted_at' }`
`audit_trail`	First or last row in `audit_trail` matching entity + action	`{ entity_type: 'application', action_type: 'create' }`
`timeline_events`	First or last row in `timeline_events` matching entity + event_type	`{ entity: 'application', event_type: 'application.shortlisted' }`

The `first: true` flag picks the earliest match (used for "first interview scheduled" etc.); default is most recent.

6.3 Defined pairs

Requisition lifecycle

- `req_raised_to_first_approval` — submitted_at → first approval audit row
- `req_raised_to_final_approval` — submitted_at → approved_at
- `req_approved_to_job_created` — approved_at → linked job.created_at

Job lifecycle

- `job_created_to_first_application` — job.created_at → first application creation audit
- `job_created_to_first_offer` — job.created_at → first `application.offered` event
- `job_created_to_first_join` — job.created_at → first `application.joined` event

Per-application stage TATs

- `applied_to_shortlisted`
- `shortlisted_to_first_interview`
- `shortlisted_to_selected`
- `selected_to_documents_cleared`
- `documents_cleared_to_ctc_sent`
- `ctc_sent_to_ctc_approved`
- `ctc_approved_to_candidate_accepted`
- `ctc_accepted_to_offer_released`
- `offer_released_to_candidate_signed`
- `offer_signed_to_joined`

End-to-end

- `applied_to_joined` — full journey

6.4 "Show working" modal

Every row in the TAT Explorer has a `Show working →` link that opens a modal containing:

1. The pair's plain-English description
2. The `from` and `to` timestamps (in IST) plus the source identifier
3. A `<details>` block with the raw row JSON used to derive each side

This makes every TAT number unambiguously auditable.

6.5 API

```
GET /api/tat/pairs?level=application
GET /api/tat/calculate?pair=...&entityId=...&entityType=...
GET /api/tat/grid?level=application&pair=...&limit=200
```

The grid endpoint returns one row per entity in scope with both the calculated duration and the source rows. Powers the explorer table.

7. Audit Deck

`/audit` (HR Admin only). Every mutation in the system writes to `audit_trail`:

```
audit_trail (
  id, action_by, action_type, entity_type, entity_id,
  before_state JSONB, after_state JSONB, metadata JSONB,
  created_at TIMESTAMP
)
```

7.1 Action types

Common values written by `logAudit()`:

- `create`, `update`, `delete`
- `approve`, `reject`, `cxo_reject`, `hr_reject`
- `blacklist`, `unblacklist`
- `assign_recruiter`, `clear_recruiter`
- `hold`, `resume`
- `move_job`, `move_talent_pool`
- `upload`, `approve_doc`, `reject_doc`
- `send_email`, `status_transition`, `schedule`

7.2 UI features

- **Thread grouping**: consecutive actions on the same entity within 30 minutes merge into one card
- **Diff pills**: red strike-through old value → emerald new value, key-prefixed
- **IST timestamps + relative time**: "Apr 25, 2026, 03:42 PM IST · 2h ago"
- **Field search**: full-text search inside `before_state` / `after_state` JSON
- **JSON export**: filtered entries exported as a single JSON file
- **Technical details** `<details>`: raw before/after/metadata JSON collapsed by default

8. Email System

8.1 Branded shell (emailBrand.js)

All transactional emails pass through `renderBrandedEmail({ title, bodyHtml, cta, context, preheader })`. Style is Goldman-recruiting-inspired:

- Single column, max 580px
- Header band with `pel.png` logo + wordmark
- Body in 15px serif-fallback sans, 1.65 line height
- Exactly one CTA per email

- Footer with IST timestamp + brand line

8.2 Transactional templates (txEmails.js)

TEMPLATE	WHEN SENT
`applicationReceivedEmail`	Self-applied candidate, immediately
`politeDeclineEmail`	HR / HOD reject with reason
`interviewScheduledEmail`	Slot confirmed
`interviewReminderEmail`	T-24h, T-30m via reminder runner
`noShowGraceEmail`	Candidate marked no-show
`ctcAcceptanceEmailV2`	CTC chain fully approved
`offerLetterReadyEmail`	Recruiter uploaded offer PDF
`offerExpiringEmail`	T-3d, T-1d before expiry
`joiningReminderEmail`	09:00 IST on joining date
`blacklistAdminAlertEmail`	Phone blacklisted, with resume attached
`requisitionRaisedAdminEmail`	Req submitted (admin alert)
`requisitionRaisedConfirmationEmail`	Req submitted (raiser confirmation)
`jobAssignedRecruiterEmail`	Job assigned to recruiter
`jobOnHoldEmail`	Job placed on hold
`shortlistInterviewerEmail`	Candidate shortlisted, interviewer notified
`ctcReviewEmail`	CTC chain step assigned to a reviewer
`joiningOutcomeEmail`	Joined / Postponed / Dropout recorded

8.3 Delivery

- Outbound via Microsoft Graph (`server/services/email.js`)
- Configured via env: `GRAPH_TENANT_ID`, `GRAPH_CLIENT_ID`, `GRAPH_CLIENT_SECRET`, `GRAPH_SENDER_EMAIL`
- Every send is logged to `email_log` with `template_id`, `to_addresses`, `cc_addresses`, `subject`, `delivery_status`
- All times rendered in IST via `formatIST()` (`Asia/Kolkata`, `Intl.DateTimeFormat`)

8.4 Reminder runner (reminders.js)

Polls `scheduled_reminders` every 30 seconds for due jobs. Handlers:

- `interview.t24h`, `interview.t30m` (candidate)

- `interview.t24h.internal` (recruiter/interviewer)
- `offer.expiring` (T-3d, T-1d)
- `joining.day` (recruiter)

Failed attempts retry up to 5 times then move to `failed`.

---

9. Database Schema Reference

All tables defined in `server/migrations/run.js`. **Idempotent** — safe to re-run. Below: every table grouped by domain.

9.1 Identity & Access

```
users (id, email UNIQUE, role, name, is_active, is_default, created_at, updated_at)
otps  (id, email, otp, expires_at, used, created_at)
notifications (id, user_email, title, message, link, read_flag, created_at)
```

9.2 Masters

```
business_units    (id, bu_name UNIQUE, bu_short_name, active_flag, …)
departments       (id, business_unit_id, department_name, active_flag)
sub_departments   (id, department_id, sub_department_name, active_flag)
locations         (id, location_name, active_flag)
phases            (id, location_id, phase_name, active_flag)
grades            (id, grade UNIQUE, active_flag)
levels            (id, level UNIQUE, active_flag)
sources           (id, source_name UNIQUE, active_flag)
reasons           (id, reason_type, reason_text, active_flag)
approvers_master  (id, employee_email, employee_name, role, business_unit_id, active_flag)
```

9.3 Hiring

```
requisitions (
  id, requisition_id UNIQUE, job_title, priority, requisition_type, job_type,
  business_unit_id, department_id, sub_department_id, location_id, phase_id,
  grade_id, level_id, experience_years, total_positions,
  start_hire_date, target_hire_date,
  status, current_approval_stage, approval_route JSONB, cxo_approval_required,
  approved_by, approved_at, approval_comments,
  submitted_by, submitted_at,
  created_by, created_at, updated_by, updated_at,
  active_flag, additional_comments, attachments JSONB
)

requisition_positions (id, requisition_id, position_type, location_id, phase_id,
  start_hire_date, target_hire_date, number_of_positions,
  backfill_employee_id, backfill_employee_name, backfill_employee_email, backfill_reason_id)

requisition_approvals (id, requisition_id, approver_email, approver_name, approver_role,
  approval_stage, status, comments, acted_by_email, acted_at, sequence)

requisition_holds (id, requisition_id, reason, placed_by_email, placed_at, lifted_at, lifted_by_

jobs (
  id, job_id UNIQUE, requisition_id, job_title, status,
  business_unit_id, department_id, sub_department_id, location_id, phase_id,
  grade_id, level_id, experience_years, job_type, requisition_type,
  job_description, additional_comments,
  compensation_currency, compensation_min, compensation_max,
  reapply_days, hiring_flow JSONB, interviewer_emails JSONB,
  publish_to_careers, allow_employee_apply, allow_employee_refer,
  number_of_positions, total_positions,
  hr_one_job_id,         -- Phase 0
  hr_one_job_ids JSONB,  -- legacy plural
  recruiter_email, secondary_recruiter_email,
  priority, active_flag,
  created_by, created_at, updated_by, updated_at
)

applications (
  id, application_id UNIQUE, ats_job_id, candidate_name, candidate_email, candidate_phone,
  candidate_age, candidate_gender, candidate_years_of_experience,
  current_organization, current_ctc, current_location, willing_to_relocate, education_level,
  source, referrer_emp_id, consultant_code, referral_flag,
  resume_path, resume_file_name, resume_flag, resume_text_preview,
  recruiter_email, secondary_recruiter_email,
  status, no_of_rounds, interviewers JSONB,
  suggested_interview_datetime1, suggested_interview_datetime2,
  interviewer_technical_score, interviewer_behavioral_score, interviewer_company_fit_score,
  interviewer_final_decision, interviewer_feedback_remarks,
  joining_date, rejected_by_email, rejection_reason, dropout_reason,
  talent_pool_only, talent_pool_expires_at,
  banned_flag, ban_scope, banned_reason,
```

```
  active_flag, is_duplicate, created_by, created_at, updated_at
)
```

9.4 Interview & Documents

```
interview_feedback (id, application_id, job_id, round_number, interviewer_email,
  scheduled_datetime, actual_datetime, status,
  technical_score, behavioral_score, company_fit_score,
  remarks, decision, rejection_reasons JSONB, attachments JSONB,
  no_show_reason, no_show_marked_by, …)

candidate_documents (id, application_id, stage, document_name, description,
  status, file_path, file_name, version, uploaded_at, uploaded_by_email,
  reviewed_by_email, reviewed_at, rejection_reason, …)

application_round_overrides (id, application_id, round_number, interviewer_emails JSONB,
  added_by_email, added_by_role, added_at, reason,
  UNIQUE(application_id, round_number))
```

9.5 CTC, Offers, Joining

```
candidate_clearance (id, application_id, status, ctc_data JSONB, ctc_text,
  primary_cleared_by, primary_cleared_at, secondary_cleared_by, secondary_cleared_at,
  hr_action, hr_action_by, hr_action_at, hr_comments,
  cxo_email, cxo_action, cxo_action_at, cxo_comments,
  aop_inline, aop_exceeded_amount, renegotiation_count)

ctc_acceptance_requests (id, application_id, requested_by, ctc_snapshot JSONB, ctc_text,
  status, responded_at, response_notes, token UNIQUE, expires_at)  -- Phase 0

ctc_approval_chain (id, application_id, step_index, role_required, assignee_email,
  assignee_name, status, acted_at, comments, ctc_text, ctc_snapshot JSONB, created_at)

offer_letters (id, application_id, file_path, file_name, uploaded_by_email, uploaded_at,
  candidate_signed_at, candidate_signature_data, candidate_signature_ip,
  candidate_decision, candidate_decision_notes, decision_at,
  validity_days, expires_at)

joining_events (id, application_id, event_type, old_date, new_date, reason,
  committed_by_email, committed_at)
```

9.6 Talent Pool & Blacklist

```
talent_pool_movements (id, application_id, candidate_email, candidate_phone,
  from_job_id, to_job_id DEFAULT 'TP-POOL', from_status,
  moved_by_email, moved_by_role, reason, moved_at)

blacklisted_phones (id, phone UNIQUE, candidate_email, candidate_name,
  reason, blacklisted_by_email, blacklisted_at, lifted_at, lifted_by_email)

-- Synthetic job seeded by the migration so applications can carry ats_job_id='TP-POOL':
INSERT INTO jobs (job_id, job_title, status, active_flag, total_positions, created_by)
SELECT 'TP-POOL', 'Talent Pool', 'open', true, 0, 'system@premierenergies.com'
WHERE NOT EXISTS (SELECT 1 FROM jobs WHERE job_id = 'TP-POOL');
```

9.7 Chat

```
candidate_chat_messages (id, application_id, sender_email, sender_role,
  body, attachment_path, attachment_name, read_by_recipient, created_at)
```

9.8 Audit, Timeline, Email log

```
audit_trail (id, action_by, action_type, entity_type, entity_id,
  before_state JSONB, after_state JSONB, metadata JSONB, created_at)

timeline_events (id, entity_type, entity_id, event_type, stage,
  actor_email, actor_role, summary, payload JSONB,
  from_state, to_state, created_at)

email_log (id, sent_by, to_addresses JSONB, cc_addresses JSONB,
  subject, body_html, context_type, context_id, sent_at,
  template_id, delivery_status, error_message)  -- last 3 are Phase 0
```

9.9 Scheduling

```
scheduled_reminders (id, kind, run_at, payload JSONB, status,
  attempts, last_error, processed_at, created_at)
```

`status` = `pending | sent | cancelled | failed`. Polled every 30s by `services/reminders.js`.

10. API Reference

All endpoints prefixed with `/api`. Auth via `Authorization: Bearer <jwt>` on protected routes.

10.1 Public

```
POST   /auth/login                        { email } → OTP sent
POST   /auth/verify-otp                   { email, otp } → { token, user }
GET    /auth/me                           → user profile
POST   /auth/logout                       → 200 + audit row
GET    /public/jobs/:jobId                → public job detail (404 if closed)
POST   /public/parse-resume               multipart, returns parsed fields
POST   /public/jobs/:jobId/apply          multipart or JSON, creates application
```

10.2 Hiring

```
GET    /requisitions                      paginated list
POST   /requisitions                      create
PUT    /requisitions/:id                  update (re-routes approvals)
GET    /requisitions/:id                  detail with approval steps
POST   /requisitions/:id/approve          CXO or HR Admin approve
POST   /requisitions/:id/reject           with reason
POST   /requisitions/:id/assign-recruiter assign recruiter
GET    /requisition-holds/:requisitionId  list holds
POST   /requisition-holds                 place hold
POST   /requisition-holds/:id/lift        lift hold

GET    /jobs                              paginated list
POST   /jobs                               create
PUT    /jobs/:id                          update (incl. hr_one_job_id, hold flow)
GET    /jobs/:id                          detail
DELETE /jobs/:id                          archive (sets status='archived')
POST   /jobs/:id/publish                  toggle publish_to_careers
GET    /jobs/:id/qr-code                  QR code image

GET    /applications                      paginated list
POST   /applications                       create
PUT    /applications/:id                  update
PUT    /applications/:id/status           transition (validates against map)
DELETE /applications/:id                  soft-delete
POST   /applications/parse-excel          multipart bulk parse
POST   /applications/upload-resume        multipart resume upload
POST   /applications/bulk-upload-excel    multipart bulk
POST   /applications/bulk-upload-resumes  multipart bulk
PUT    /applications/:id/interview-plan   update no_of_rounds + interviewers
POST   /applications/:id/move-stage       legacy stage move
```

10.3 Triage & Blacklist

```
POST   /triage/:id/shortlist              { no_of_rounds, interviewers_per_round }
POST   /triage/:id/move-to-job            { target_job_id }
POST   /triage/:id/move-to-talent-pool    { reason }
POST   /triage/:id/hr-reject              { reason }
GET    /triage/jobs/searchable            ?q= → live search

GET    /blacklist                         list bans
POST   /blacklist                         { application_id, reason }
POST   /blacklist/check                   { phone } → { blacklisted, reason }
DELETE /blacklist/:phone                  lift (admin only)
```

10.4 Interviews

```
GET    /interviews                         per-user assigned candidates
PUT    /interviews/:id/suggest-slots      interviewer suggests two
PUT    /interviews/:id/confirm-slot       recruiter confirms or overrides
PUT    /interviews/:id/feedback           scores + decision
PUT    /interviews/:id/mark-no-show       with 10-min lockout
PUT    /interviews/:id/request-additional-rounds  with new interviewers + slots
```

10.5 Documents & Chat

```
GET    /candidate-portal/me               candidate's own application + docs
POST   /candidate-portal/documents/:docId/upload  multipart
GET    /candidate-portal/review-queue     HR review queue (recruiter-scoped)
POST   /candidate-portal/documents/:docId/review  { decision, review_notes }
POST   /candidate-portal/:applicationId/ctc-request  HR sends CTC
POST   /candidate-portal/ctc-request/:id/respond  candidate accepts / declines / renegotiates

GET    /chat/:applicationId/thread         recruiter view
POST   /chat/:applicationId/send           recruiter send (multipart)
GET    /chat/me/thread                     candidate view
POST   /chat/me/send                       candidate send (multipart)
```

10.6 CTC chain & Offers

```
POST   /ctc-chain/:applicationId/start     { ctc_text, ctc_snapshot, secondary_recruiter_email,
GET    /ctc-chain/:applicationId/chain     list steps
POST   /ctc-chain/:applicationId/act       { decision: approved|rejected|renegotiate, comments
POST   /ctc-chain/me/accept                candidate accepts CTC

POST   /offers/:applicationId/upload       multipart, recruiter uploads PDF
GET    /offers/me/current                  candidate's active offer
POST   /offers/me/sign                     { signature_data, decision, decision_notes }
POST   /offers/:applicationId/joining      { joining_date, reason, event_type }
POST   /offers/:applicationId/joining-outcome  { outcome: joined|postpone|dropout, reason }
```

10.7 Analytics

```
GET    /mis/headline                        org-level KPIs
GET    /mis/funnel                          stage distribution
GET    /mis/monthly-offers                  recruiter × month
GET    /mis/recruiter-sourcing              recruiter × source
GET    /mis/time-to-fill                    job-level fill TAT
GET    /mis/source-mix                      source-wise totals
GET    /mis/department-health               per-department metrics
GET    /mis/backouts-summary                rejected/dropout reasons
GET    /mis/raw-export                      one row per application — every column
GET    /mis/jobs-export?include_applicants= one row per job (or job × app)
GET    /mis/detailed-open-positions         workbook-style aging report
POST   /mis/assistant                       { question } → AI answer

GET    /tat/pairs?level=                    catalog
GET    /tat/calculate                       single calc
GET    /tat/grid?level=&pair=               table for explorer

GET    /audit                               list with filters
GET    /audit/stats                         KPI summary
GET    /audit/export                        filtered JSON export

GET    /timeline?entity_type=&entity_id=    polymorphic timeline
```

10.8 Filters available on MIS

Every MIS endpoint shares `buildFilters()`:

```
date_from, date_to             ISO date strings, inclusive
recruiter_email | recruiter
business_unit_id, department_id, location_id, phase_id
hr_one_job_id                  exact match
status                          string or array
source                          string
talent_pool                    'include' (default) | 'exclude' | 'only'
```

11. Operations & Setup

11.1 First-time setup

```
# 1. Install all deps
npm run install:all

# 2. Set up Ollama for resume parsing (optional but recommended)
npm run setup:ai

# 3. Create .env in project root with:
#    PG_URL=postgres://user:pass@host/db?sslmode=require
#    JWT_SECRET=<random 64 hex>
#    APP_URL=https://ats.premierenergies.com
#    GRAPH_TENANT_ID=...
#    GRAPH_CLIENT_ID=...
#    GRAPH_CLIENT_SECRET=...
#    GRAPH_SENDER_EMAIL=spot@premierenergies.com

# 4. Apply schema
npm run migrate

# 5. Seed defaults
npm run seed

# 6. Boot
npm run dev
```

11.2 Daily operations

TASK	HOW
Add a recruiter	`/users` → + Add User
Configure org chart	`/masters` → BU / Dept / Location tabs
Define approver mapping	`/masters` → Approvers Master
Open a job	Approve a requisition → "Create Job"
Bulk-upload candidates	`/jobs/:id` → "+ Add Candidate" → "Bulk from Excel" or "Bulk from Resumes"
Ban a phone	Candidate row → Triage → Blacklist (with reason)
Lift a ban	`/audit` shows blacklist row; HR Admin uses `DELETE /api/blacklist/:phone`
Move multiple candidates to TP	One at a time via Triage menu (intentional, audited)
Run a TAT analysis	`/tat` → pick level + pair → Show working on any row
Export raw data	`/mis` → Raw Export tab → Export CSV
Review document loop	`/hr/document-queue`

11.3 Demo data

HR Admin can:

- **Seed demo** — populates BUs, jobs, applicants for screenshots/demos
- **Clear demo** — wipes all demo-tagged data
- **Run full demo** — animates a candidate through the entire flow

11.4 Logs & monitoring

- `email_log` — every outbound email with delivery status
- `audit_trail` — every mutation
- `timeline_events` — polymorphic event log for entity timelines
- `scheduled_reminders` — pending + sent + failed reminders

12. Troubleshooting

SYMPTOM	LIKELY CAUSE	FIX
`npm run migrate` fails with NOT NULL on `created_by`	Old jobs row missing `system@…`	Migration now seeds it; re-run after pulling latest
Public apply returns 410 "no longer accepting"	Job status is `closed/archived/on_hold/filled/cancelled`	Check job status on `/jobs`
Public apply returns 403 "could not be processed"	Phone matches an active blacklist entry	Lift via `/audit` → DELETE `/api/blacklist/:phone`
Candidate self-apply gets "duplicate" message	Same email already applied to this exact job	Recruiter checks talent pool; can re-engage
Offer signature 410 expired	More than `validity_days` since upload	Recruiter re-uploads offer letter
No-show button returns 425	Less than 10 minutes past scheduled time	Wait for the lockout to clear
Reminder runner not firing	`index.js` not invoking `remindersService.start()`	Verify import + call in server bootstrap
Status transition rejected	Move violates the transition map	See §5 Application Status Map
TAT shows "missing" for "from" or "to"	The required event hasn't been emitted yet	Confirm the underlying transition wrote a `timeline_events` or `audit_trail` row
Email log shows `delivery_status='failed'`	Graph credentials invalid or sender not licensed	Verify `GRAPH_*` env vars
MIS shows wrong recruiter rows	GROUP BY mismatch between SELECT and aggregation	Already fixed in `routes/mis.js` — uses `COALESCE(...)` consistently

Appendix A — File Map

Server

FILE	PURPOSE
`server/index.js`	Boot, routes mount, static brand assets, reminder runner start
`server/middleware/auth.js`	JWT verify, `requireRole` factory
`server/services/email.js`	Graph send + in-app notification
`server/services/emailBrand.js`	Branded email shell + IST helpers
`server/services/txEmails.js`	14 transactional templates
`server/services/timeline.js`	`logTimeline`, polymorphic event service
`server/services/audit.js`	`logAudit`
`server/services/tat.js`	TAT pair definitions + resolver
`server/services/reminders.js`	Setinterval-based reminder runner
`server/services/interviewWorkflow.js`	`assertHrManagedTransition`, `HR_MANAGED_TRANSITIONS`, `ALL_APPLICATION_STATUSES`
`server/services/requisitionApproval.js`	Resolve type → status, build approval route
`server/services/ai.js`	Ollama-backed resume parsing + assistant
`server/routes/triage.js`	Move-to-pool / move-to-job / shortlist / hr-reject
`server/routes/blacklist.js`	Phone ban + admin alerts
`server/routes/chat.js`	Two-sided message thread
`server/routes/offers.js`	Offer upload, signature, joining outcome
`server/routes/ctcChain.js`	Recruiter1→Recruiter2→HR Admin→Approver chain
`server/routes/tat.js`	`/pairs`, `/calculate`, `/grid`

Client

FILE	PURPOSE
`client/src/styles/index.css`	Base tokens — Workday/Greenhouse-flavoured
`client/src/styles/v2.css`	Premium animations, gradients, primitives layer
`client/src/components/ui/v2.jsx`	Hero, KPI, StatusPillV2, TimelineRail, WorkingModal
`client/src/components/ui/v2plus.jsx`	AuroraHero, LoudKPI, RingProgress, Sparkline, Marquee, AnimatedNumber, MagneticButton
`client/src/components/TriageMenu.jsx`	5-modal triage dispatcher
`client/src/components/SignaturePad.jsx`	Canvas signature capture
`client/src/components/OfferSignaturePanel.jsx`	Candidate offer view + sign
`client/src/components/CandidateChatPanel.jsx`	Candidate↔Recruiter chat
`client/src/components/CtcChainModal.jsx`	HR-side CTC chain UI
`client/src/pages/TatExplorer.jsx`	TAT page with Show working modal
`client/src/pages/AuditDeck.jsx`	Audit deck with thread grouping + diff pills

End of guide. Questions or gaps? Open `/audit` to see exactly what happened — every action ever taken in the system is documented there in plain English with IST timestamps.